Signet 01

Jim Hannon-Tan, 2026

925 sterling silver, resonant planar antenna, secure cryptographic element
Foundation Edition · 10 numbered pieces worldwide

The premise

Signet rings historically authenticated documents through a personal mark. Signet 01 establishes this principle within digital architecture, where autonomous software and synthetic agents increasingly act on our behalf.

The instrument binds a hardware-isolated signing key to a deliberate physical gesture. Touching the ring to a host device creates a sovereign, unforgeable mandate.

The architecture

The chassis is investment-cast in 925 sterling silver and hand-finished. The continuous twelve-turn square spiral on its face serves simultaneously as the identifying hallmark and the functional radio antenna.

Brought close to a host interface, the coil receives transient energy and data via NFC. That inducted energy powers the secure element directly. The instrument requires no internal battery, charging, or electrochemical maintenance.

Cryptographic intent

The host system constructs the directive as an EIP-712 structured data payload and derives its digest. The internal secure element signs that digest using ECDSA and a private key generated and permanently isolated on-silicon.

The resulting cryptographic seal verifies against open standards using the ring's public key. Any post-signature modification invalidates the digest immediately. The instrument does not delegate authority; it asserts direct physical presence.

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Technical specifications

Antenna	Integrated 12-turn planar square spiral · 13.56 MHz
Secure element	Arx HaLo passive NFC
Signing algorithm	ECDSA / secp256k1 standard
Data standard	EIP-712 typed structured data
Identity primitive	LUKSO Universal Profile integration (ERC-725 / ERC-734)

Provenance & The Register

Signet 01 interfaces with LUKSO Universal Profiles, allowing the ring's signing key to act as a sovereign controller with definable on-chain permissions. Key permissions can be updated or revoked without altering the underlying profile.

The Register records each numbered piece's metallurgical origin, casting date, ownership lineage, and authorised key transitions.

The movement & servicing

Engineered for indefinite longevity, the ring's cryptographic calibre is designed for studio replacement. The solid silver chassis and numbered hallmark are preserved, refinished, and fitted with a new calibre should cryptographic or silicon standards evolve. Every authorised calibre replacement is certified and recorded within The Register.

Concept & industrial design

Jim Hannon-Tan is an industrial designer whose work spans everyday objects, medical technology, and digital instruments. His studio practice incorporates extensive international experience across Milan, Amsterdam, and London, and is currently based in Adelaide, Australia.

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Precursor: Near Field Circuit

Signet 01 grew from Near Field Circuit, Genesis Edition (2025): ten seals CNC-machined from solid brass, copper-plated and hand-assembled, each carrying an Arx HaLo chip bonded to an LSP8 token on LUKSO. A tap recorded presence and ownership to the holder's Universal Profile. All ten were sold and are held by collectors around the world.

nearfieldcircuit.com

Development & support

Near Field Circuit was supported by the LUKSO Grants Program. That work adapted Arx's software libraries to LUKSO standards, connecting HaLo NFC chips with LSP8 digital assets, and Signet 01 builds directly on it.

The project has also received ANFF-C Gate 1 funding to support its development and commercialisation.

Collaborators

Blockchain development	Aureliano Alarcón Smith
Mechanical engineering	Varghese Liju
Electronic engineering	Aaron Mohtar
RF and antenna consultancy	Dr Shengjian Chen, Flinders University
Casting	Morris & Watson
Bench work	Shannon Cornish Artisan Jewellers
Vitreous enamel R&D	JamFactory, Adelaide

Special thanks to Cameron Robertson, Founder of Arx, and Fabian Vogelsteller, Co-founder of LUKSO, for their support.

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Allocations & inquiries

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